

AYUSH BHALERAO

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SUMMARY

Software Engineer specializing in scalable backend systems, AI integration, and modern Fullstack web infrastructure. Experienced in building production SaaS products, semantic similarity & vectorDBs products, high-concurrency APIs, moderation pipelines, and implementing cost-effective serverless architectures.

EXPERIENCE

Founding Engineer

March – June

AlphaJEE(Acquired)

Remote

- Led & shipped several projects including an **AI enabled Adaptive Testing platform**, **client-side parsers**, **data analytics dashboards**, **social forum**, and **ml based moderation pipelines** in time-constrained environment.
- Developed the full rate-to-score pipeline with semantic similarity using 2PL-IRT for an **AI enabled Adaptive Question Solving platform**.made per student knowledge graphs with Next.js, Go (gorilla/mux), and PostgreSQL(w/ pgvector).
- Built a **JEE Mains response sheet checker** in Next.js with a client-side scraper/parser, serving **28k+ response sheet evaluations**.
- Built a **multi-stage moderation pipeline** (SBERT, NudeNet, Gemini API) on **GCP Cloud Run** with FASTAPI to serve it as an endpoint.
- Migrated core infrastructure from Supabase to **Cloud SQL (PostgreSQL)**, improving scalability for high-concurrency workloads.

PROJECTS

FileSense – Semantic File Organizer | Python, SBERT, FAISS, LLM API | [GitHub](#) | [demo](#)

Dec 2025

- Built a **local-first semantic file organizer** using sentence embeddings(SBERT) and FAISS vector search to classify documents by content, with Gemini API fallback to auto-generate new categories on unseen file contents and categories.
- Designed a self-updating pipeline (rebuilds FAISS index on new labels) achieving **89% categorization rate**; implemented an **epsilon-greedy RL agent** for adaptive folder-assignment policy over static thresholds which also helped in reducing API calls.
- Compared [Natural Language vs Keyword](#) descriptions for categories, concluded Keyword descriptions are superior - [planning](#) to Super fine-tune gemma-4b due to recent gemini-api ratelimits.

NoCursor – CLI code editor | Python, llama.cpp, git | [GitHub](#) | [demo](#)

Jan 2026

- A **completely local, CLI-first, determinstic** coding assistant with context-aware fix, plan, apply modes.
- Uses **llama.cpp** with llama-server serving openweight models, tested with DeepSeek-R1-Distill-Qwen-1.5B & Qwen2.5-Coder-q4-3B, deepseek-r1-1.5B performing arguably better for its size.
- Implemented file **context-awareness**, **basic project awareness**, **git diffs** (partly supported with vscode).

BetJEE – Prediction Market | REST APIs, NextJS, PostgreSQL | [Blog](#) | [GitHub](#)

Feb 2026

- Built a virtual prediction market withing 3 days which handled **2300+ database users and 1.2M requests in 48 hours** via Supabase Edge Functions.
- Developed **recursive stop-loss systems** in PostgreSQL RPCs, proofing to avoid race conditions and automated **anti-bot IP blacklisting**.

[→ More projects on my portfolio](#)

TECHNICAL SKILLS

Languages: Python, Go, C, SQL (PostgreSQL), JavaScript/TypeScript

ML & Data: FAISS index, sentence embeddings(SBERT), pgvector, LLaMA.cpp, Pandas, NumPy, Regex, scikit-learn

FullStack & AI integration: Gemini API, Grok API, Openrouter

Backend & Tools: REST APIs, WebSockets, Cloudflare Workers, FastAPI, Supabase, Google Cloud Platform (GCP), Node.js, Git/GitHub, Linux

EDUCATION

City Pride School

AISSE – CBSE (XII), Grade: 87.2%

2026

Pune

Dr. D.Y. Patil Public School

AISSE – CBSE (X), Grade: 93%

2024

Pune